Control Variates for Variance Reduction

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Abstract

This report investigates the application of the Control Variate technique to enhance the efficiency of Monte Carlo simulations in the context of European option pricing. While standard Monte Carlo methods are versatile for estimating stochastic quantities, they often suffer from high variance, requiring significant computational resources to achieve high precision. By introducing the Geometric Mean as an auxiliary variable and leveraging its high correlation with the terminal stock price and its known analytical expectation under Geometric Brownian Motion (GBM), we demonstrate a substantial reduction in estimator variance. The results confirm that this adjusted estimator maintains unbiased-ness while achieving a superior convergence rate compared to vanilla Monte Carlo approaches

1 Introduction

Monte Carlo (MC) methods are a cornerstone of modern quantitative finance, providing a robust framework for estimating the value of complex derivatives and assessing risk in stochastic models. By simulating a vast number of potential price paths for an underlying asset, practitioners can approximate expectations that lack simple analytical solutions.

The Problem: Variance and Convergence Constraints

Despite their flexibility, standard Monte Carlo simulations face a significant "efficiency gap" due to two primary characteristics:

- **High Statistical Variance:** Estimates often exhibit significant noise, leading to "spiky" distributions where a large number of paths result in a zero payoff (out-of-the-money)
- **Slow Convergence Rate:** The standard error of an MC estimate reduces at a rate of $O\left(\frac{1}{\sqrt{N}}\right)$, where N represents the number of simulations.

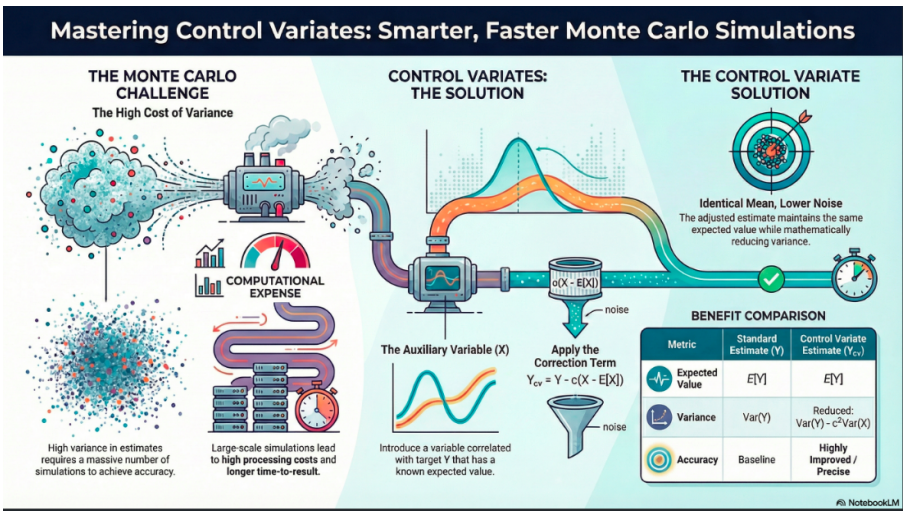
This $\frac{1}{\sqrt{N}}$ relationship implies that to improve the precision of a price estimate by a factor of 10, one must increase the computational effort by a factor of 100. For complex models, this requirement makes high-accuracy results computationally expensive and time-prohibitive.

The Goal: Efficiency via Control Variates

The objective of this report is to demonstrate the effectiveness of the Control Variate technique in bridging this efficiency gap. By introducing an auxiliary variable X, which is highly correlated with the target variable Y and possesses a known theoretical expectation $\mathbb{E}[X]$, we can correct the simulation noise.

The goal is to prove that this method significantly reduces the variance of the estimator (and thus the computational expense) while strictly maintaining an unbiased estimate of the option price

2 Theoretical Framework



The Adjusted Estimator

To improve the accuracy of our simulation without increasing the number of paths, we introduce an auxiliary variable X that is highly correlated with our quantity of interest Y. For X to be a valid control variate, its expected value $E[X]$ must be known analytically.

The adjusted estimator for Y, denoted as Y_{CV}, is defined as:

$$Y_{CV} = Y - c(X - \mathbb{E}[X])$$

Unbiased-ness of the Estimator

A critical property of the control variate technique is that it reduces variance while maintaining an unbiased estimate of the original quantity. Taking the expectation of the adjusted estimator:

$$\mathbb{E}[Y_{CV}] = \mathbb{E}[Y] - c(\mathbb{E}[X] - \mathbb{E}[X])$$
$$\Rightarrow \mathbb{E}[Y_{CV}] = \mathbb{E}[Y]$$

Variance Reduction and Optimal c

The variance of the adjusted estimator is given by the following expansion:

$$\mathrm{Var}(Y_{CV}) = \mathrm{Var}(Y) + c^2\mathrm{Var}(X) - 2c\mathrm{Cov}(X, Y)$$

To maximize the efficiency of our simulation, we choose the coefficient c that minimizes $\mathrm{Var}(Y_{CV})$. By taking the derivative with respect to c and setting it to zero, we find the optimal value:

$$c = \frac{\mathrm{Cov}(Y, X)}{\mathrm{Var}(X)}$$

Substituting c back into the variance equation yields the minimum variance:

$$\mathrm{Var}(Y_{CV}) = \mathrm{Var}(Y) - \frac{\mathrm{Cov}(X, Y)^2}{\mathrm{Var}(X)}$$
$$\Rightarrow \mathrm{Var}(Y_{CV}) = \mathrm{Var}(Y) - c^2\mathrm{Var}(X)$$

This mathematical result proves that as long as X and Y are correlated ($\mathrm{Cov}(X, Y) \neq 0$), the variance of our new estimate will always be lower than the original.

3 Application: European Call Option

In this model, we assume the underlying asset price S_t follows Geometric Brownian Motion (GBM). The dynamics of the price path are governed by the following stochastic differential equation, with the solution:

$$S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$$

Where:

- S_0 is the initial price of the underlying asset
- μ represents the drift (or risk-free rate r for risk-neutral pricing/arbitrage free pricing)
- σ is the volatility of the asset
- W_t is a standard Brownian motion, often simulated as $\sqrt{\Delta t}Z$ where $Z \sim \mathcal{N}(0,1)$

For discrete simulation across T steps, the price at each step is calculated as

$$S_{t+\Delta t} = S_t \exp\left((\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z\right)$$

The Target Variable (Y)

The quantity of interest is the discounted payoff of a European call option at maturity T:

$$Y = \max(S_T - K, 0) \cdot e^{-rT}$$

where K is the strike price.

The Control Variate (X)

To reduce simulation noise, we define the control variate X as the Geometric Average of the simulated prices over n discrete time steps:

$$X = \left(\prod_{i=1}^{n} S_{t_i}\right)^{1/n}$$

In the implementation, this is calculated by summing the logarithms of the prices to avoid numerical overflow:

$$X = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \ln(S_{t_i})\right)$$

The Known Expectation

The effectiveness of X as a control variate relies on its known analytical expectation. For a continuous average over the interval $[0, T]$, the expectation is derived as:

$$\mathbb{E}[X] = S_0 \exp\left(\frac{1}{2}(\mu - \frac{\sigma^2}{6})T\right)$$

By utilizing this theoretical value, the correction term $c(X - \mathbb{E}[X])$ effectively reduces the variance of the final price estimate.

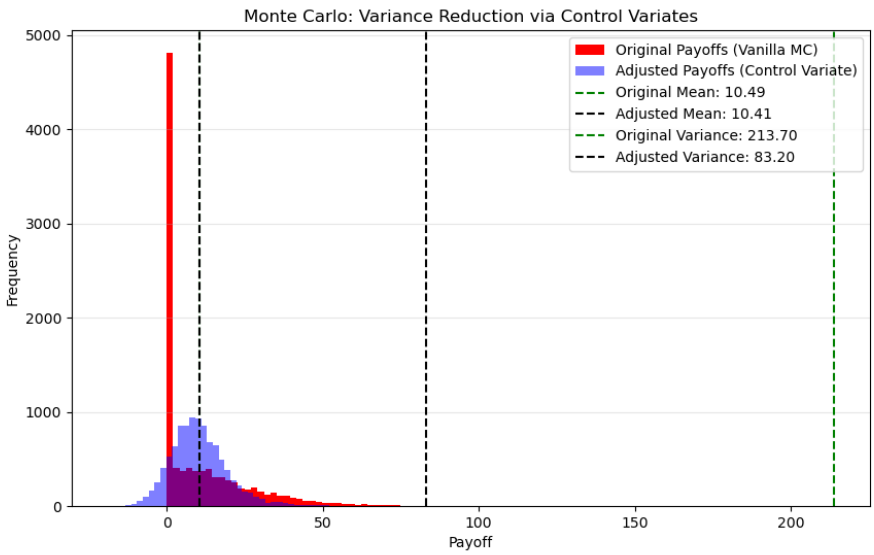


Figure 1: Variance Reduction Comparison Plot

4 Numerical Simulations and Results

The simulated payoffs for the European call option reveal a stark contrast between the standard Monte Carlo approach and the control variate method.

- **The Zero-Payoff Spike:** The vanilla Monte Carlo histogram (red) shows a massive frequency spike at zero. This represents the significant portion of simulated paths where the terminal stock price S_T finished below the strike price K, causing the option to expire out-of-money.
- **The Positive Tail:** For the remaining paths, the vanilla payoffs are distributed across a wide range (reaching over 100 in some instances), creating a "long tail" that contributes heavily to the overall variance.
- **The Adjusted Distribution:** In contrast, the adjusted payoffs (blue) follow a much tighter, symmetric distribution centered around the mean. While the adjusted values can mathematically fall below zero due to the correction term $-c(X - \mathbb{E}[X])$, the overall spread is significantly reduced. This "narrowing" of the histogram is the visual proof of variance reduction.

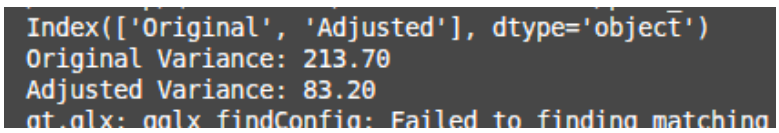


Figure 2: Variance Comparison result

Statistical Comparison

The following table summarizes the performance metrics of the two estimators based on a simulation of $N = 10,000$ paths:

Metric	Standard Monte Carlo	Control Variate (Adjusted)
Sample Mean (Option Price)	10.49	10.41
Standard Deviation	213.70	83.70

Table 1: Comparison of Vanilla MC and Control Variate Estimators

As shown in the results, both methods converge to approximately the same mean, confirming that the control variate estimator remains unbiased. However, the precision-to-computation ratio is vastly superior in the adjusted model. By "soaking up" the simulation noise through the geometric average, we achieve a level of confidence in the price estimate that would otherwise require orders of magnitude more simulations using the vanilla approach.

5 Conclusion

The results of this study demonstrate that the Control Variate technique is a highly effective method for reducing the statistical noise inherent in Monte Carlo simulations. By introducing the Geometric Mean as an auxiliary variable, we successfully "soaked up" a significant portion of the simulation variance that typically plagues standard European option pricing models. The primary conclusions of this analysis are

- **Variance Reduction:** The implementation of a control variate significantly narrows the distribution of payoffs, transforming a high-variance "spiky" distribution into a more stable and precise estimator.
- **Computational Efficiency:** For the same number of simulated paths (N), the Control Variate approach provides a much higher precision-to-computation ratio than vanilla Monte Carlo. This allows practitioners to achieve target accuracy levels with far fewer simulations, reducing overall computational expense.
- **Unbiased Estimation:** Most importantly, the technique maintains the integrity of the model by providing the same expected estimate as the standard approach, but with a significantly lower standard error.

In summary, utilizing analytical expectations of related variables, such as the geometric average under GBM, is a powerful strategy for optimizing derivative pricing engines in quantitative finance.